Akoizo Original Practice - 2027 Rotation Form 2

Division C | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (400 points) Decode the timed-base Aristocrat: RAFRV FUFSP CFGGFS KFMXSF DRRFEGHQT D ZXCNGHXQ

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet: DKRYFMTAHOVCJQXELSZGNUBIPW. Each row means plaintext letter -> ciphertext letter. This is a supplied one-to-one monoalphabetic substitution; no letter maps to itself. Timing is practiced separately; this worksheet awards the base score only.

Answer: _____
_____
_____

2. (400 points) Decode the Patristocrat: RNARF AXAPO GAUUA PEATK PASRR ALUCJ ISQKG WUCKJ

Spaces were removed and groups of five added for display.
Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet (K2, keyed and cyclically shifted): SERVATINCDFGHJKLMPQUWXYZOB

Answer: _____
_____
_____

3. (400 points) Decode this Spanish Xenocrypt: GKQ VSUKQ SOWVSJ S LAJQSP

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet (K2, keyed and cyclically shifted): SERVATINCDFGHJKLMPQUWXYZOB. Plaintext is Spanish; no accent marks are encoded.

Answer: _____
_____
_____

4. (400 points) Decode this Baconian message: baabb aabaa baaba baabb abaaa aaabb aabaa aaaaa baaba

Use the supplied modern 26-letter mapping:
A=aaaaa; B=aaaab; C=aaaba; D=aaabb; E=aabaa; F=aabab; G=aabba; H=aabbb; I=abaaa; J=abaab; K=ababa; L=ababb; M=abbaa; N=abbab; O=abbba; P=abbbb; Q=baaaa; R=baaab; S=baaba; T=baabb; U=babaa; V=babab; W=babba; X=babbb; Y=bbaaa; Z=bbaab. Spaces separate encoded letters; restore word boundaries from meaning. Crib READ/TEST/CHECK/KEEP, as applicable: the plaintext begins with TEST.

Answer: _____
_____
_____

5. (400 points) Decode this Nihilist message: 57 24 68 26 47 24 59

Use key DATA, repeated. I/J share a cell. Coordinates are row then column. Encryption adds plaintext coordinate plus key coordinate as ordinary integers.
1: A B C D E
2: F G H I K
3: L M N O P
4: Q R S T U
5: V W X Y Z

Answer: _____
_____
_____

6. (400 points) Decode this 5x5 Checkerboard message: CZ AV AX BZ CW EY AW CY EX EW BY DY BX BV BY EV AZ AY CY EZ AZ CX DV DZ AV DW DY DX

Rows are labeled ABCDE; columns VWXYZ. Read each pair row first. Polybius grid rows in order:
1: A B C D E
2: F G H I K
3: L M N O P
4: Q R S T U
5: V W X Y Z. All row and column labels occur in the ciphertext.

Answer: _____
_____
_____

7. (400 points) Solve this complete-columnar transposition: AUNCINTS SCTENTMO RNEOSTHX TOSDSEED

Four columns were filled left-to-right by rows; whole columns were read in an unknown order. Trailing X characters pad the last row. The plaintext begins with crib STAR at its first four positions. Recover the complete plaintext including padding X characters. There are no omitted cells.

Answer: _____
_____
_____

8. (2 points) A Homophonic Cipher keyword has four letters. Which property must those letters satisfy under the 2027 C rules?

Answer: _____
_____
_____

9. (2 points) A timed Aristocrat is solved correctly in 180 seconds. Compute the timing bonus using 2(600-t).

Answer: _____
_____
_____

10. (2 points) A 400-point plaintext answer has five character errors. Two errors are free and each further error costs 100 points. Find its score.

Answer: _____
_____
_____

11. (2 points) A 500-point keyword answer has three character errors. No keyword errors are free and each error costs 100 points. Find its score.

Answer: _____
_____
_____

12. (2 points) A proposed Aristocrat alphabet maps plaintext A to ciphertext A. Why is it invalid under the 2027 C rules?

Answer: _____
_____
_____

13. (2 points) A Fractionated Morse problem supplies a crib but does not identify its ciphertext placement. What information is missing under the 2027 C rules?

Answer: _____
_____
_____

14. (400 points) Decode this Fractionated Morse message: DUFJYTVLAIDGHG

A Morse letter separator is x and a word separator is xx; trailing x is padding. The plaintext begins with READ at its first four letters. Supplied fractionation table: A=...; B=..-; C=..x; D=.-.; E=.--; F=.-x; G=.x.; H=.x-; I=.xx; J=-..; K=-.-; L=-.x; M=--.; N=---; O=--x; P=-x.; Q=-x-; R=-xx; S=x..; T=x.-; U=x.x; V=x-.; W=x--; X=x-x; Y=xx.; Z=xx-. Morse letters: A=.-; B=-...; C=-.-.; D=-..; E=.; F=..-.; G=--.; H=....; I=..; J=.---; K=-.-; L=.-..; M=--; N=-.; O=---; P=.--.; Q=--.-; R=.-.; S=...; T=-; U=..-; V=...-; W=.--; X=-..-; Y=-.--; Z=--...

Answer: _____
_____
_____

15. (400 points) Decode this Porta message: YVVRQDTJGPEM

Given keyword SCIENCE, repeated over letters only. A=0..Z=25. Each key pair selects a reciprocal substitution row. Upper alphabet for every row is ABCDEFGHIJKLMNOPQRSTUVWXYZ; lower rows: AB=NOPQRSTUVWXYZABCDEFGHIJKLM; CD=OPQRSTUVWXYZNMABCDEFGHIJKL; EF=PQRSTUVWXYZNOLMABCDEFGHIJK; GH=QRSTUVWXYZNOPKLMABCDEFGHIJ; IJ=RSTUVWXYZNOPQJKLMABCDEFGHI; KL=STUVWXYZNOPQRIJKLMABCDEFGH; MN=TUVWXYZNOPQRSHIJKLMABCDEFG; OP=UVWXYZNOPQRSTGHIJKLMABCDEF; QR=VWXYZNOPQRSTUFGHIJKLMABCDE; ST=WXYZNOPQRSTUVEFGHIJKLMABCD; UV=XYZNOPQRSTUVWDEFGHIJKLMABC; WX=YZNOPQRSTUVWXCDEFGHIJKLMAB; YZ=ZNOPQRSTUVWXYBCDEFGHIJKLMA.

Answer: _____
_____
_____

16. (2 points) How do keyword-answer scoring and plaintext-answer scoring differ in their free-error allowances?

Answer: _____
_____
_____

17. (2 points) Why must a monoalphabetic substitution use a globally consistent mapping?

Answer: _____
_____
_____

18. (2 points) A timed puzzle is first solved after 650 seconds. What timing bonus remains?

Answer: _____
_____
_____

19. (2 points) Two special-bonus questions are solved with no penalty points. What aggregate special bonus does the 2027 C schedule award?

Answer: _____
_____
_____

20. (2 points) Which permitted monoalphabetic format removes word spaces and displays ciphertext in groups of five?

Answer: _____
_____
_____

21. (200 points) Use Figure F1. Decode Message 1 using the row-first pairs and the pictured grid. I/J share a cell.

Given row/column labels and full Polybius key

	V	W	X	Y	Z
A	D	E	F	G	H
B	I	K	L	M	N
C	O	P	Q	R	S
D	T	U	V	W	X
E	Y	Z	A	B	C

Message 1

DX AW CY BV AX EV AW EX EZ AZ BY CV AV AW BX DY BV DV AZ CZ DV CY CV BZ AY AW DX BV AV AW BZ EZ AW

Message 2

EZ AZ AW EZ BW AW DX AW CY EV CZ EV BY EY CV BX EX BZ AV CY AW EZ CV CY AV CY AW CZ DW BX DV CZ

Message 3

EZ CV BY CW EX CY AW DV AZ AW CV EY CZ AW CY DX AW AV AV EX DV EX DY BV DV AZ CW CY AW AV BV EZ DV BV CV BZ CZ

F1. Checkerboard coordinate grid (original synthetic teaching figure)

Answer: _____
_____
_____

22. (200 points) Use Figure F1. Decode Message 2 using the row-first pairs and the pictured grid. I/J share a cell.

Refer to F1. Checkerboard coordinate grid (original synthetic teaching figure) shown with the first question in this figure group.

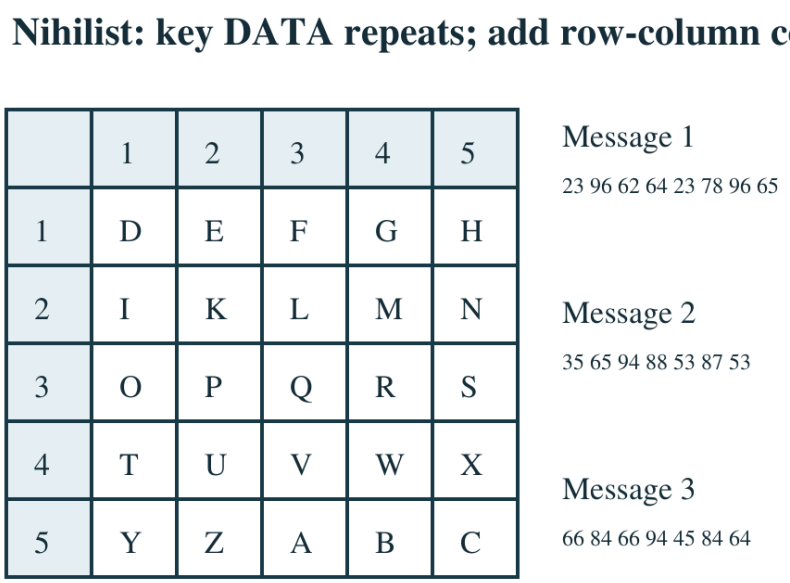
Answer: _____
_____
_____

23. (200 points) Use Figure F1. Decode Message 3 using the row-first pairs and the pictured grid. I/J share a cell.

Refer to F1. Checkerboard coordinate grid (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (200 points) Use Figure F2. Decode Nihilist Message 1. Use the pictured grid and the given repeating key DATA; subtract its coordinate values as integers.



F2. Nihilist cipher station (original synthetic teaching figure)

Answer: _____
_____
_____

25. (200 points) Use Figure F2. Decode Nihilist Message 2. Use the pictured grid and the given repeating key DATA; subtract its coordinate values as integers.

Refer to F2. Nihilist cipher station (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

26. (200 points) Use Figure F2. Decode Nihilist Message 3. Use the pictured grid and the given repeating key DATA; subtract its coordinate values as integers.

Refer to F2. Nihilist cipher station (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (200 points) Use Figure F3. Decode M1 with the pictured Porta chart. Use the keyword printed above the chart, repeated across letters only.

Given-key Porta; repeat keyword over letters only

Upper alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ; key CONTROL

AB: NOPQRSTUVWXYZABCDEFGHIJKLM
CD: OPQRSTUVWXYZNMABCDEFGHIJKL
EF: PQRSTUVWXYZNOLMABCDEFGHIJK
GH: QRSTUVWXYZNOPKLMABCDEFGHIJ
IJ: RSTUVWXYZNOPQJKLMABCDEFGHI
KL: STUVWXYZNOPQRIJKLMABCDEFGH
MN: TUVWXYZNOPQRSHIJKLMABCDEFG
OP: UVWXYZNOPQRSTGHIJKLMABCDEF
QR: VWXYZNOPQRSTUFGHIJKLMABCDE
ST: WXYZNOPQRSTUVEFGHIJKLMABCD
UV: XYZNOPQRSTUVWDEFGHIJKLMABC
WX: YZNOPQRSTUVWXCDEFGHIJKLMAB
YZ: ZNOPQRSTUVWXYBCDEFGHIJKLMA

M1: QHSGVKWSBOZZGUS

M2: DYVFJXBVYJWLMWDG

M3: QOXYSYDSKFIZLCZM

F3. Porta reciprocal alphabet chart (original synthetic teaching figure)

Answer: _____
_____
_____

28. (200 points) Use Figure F3. Decode M2 with the pictured Porta chart. Use the keyword printed above the chart, repeated across letters only.

Refer to F3. Porta reciprocal alphabet chart (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

29. (200 points) Use Figure F3. Decode M3 with the pictured Porta chart. Use the keyword printed above the chart, repeated across letters only.

Refer to F3. Porta reciprocal alphabet chart (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (200 points) Use Figure F4. Decode Message 1 with the pictured Baconian chart. Cipher groups encode letters. The plaintext starts with READ; restore word spaces.

Modern26-letter mapping; groups encode individual letters

A = aaaaa
B = aaaab
C = aaaba
D = aaabb
E = aabaa
F = aabab
G = aabba
H = aabbb
I = abaaa
J = abaab
K = ababa
L = ababb
M = abbaa
N = abbab
O = abbba
P = abbbb
Q = baaaa
R = baaab
S = baaba
T = baabb
U = babaa
V = babab
W = babba
X = babbb
Y = bbaaa
Z = bbaab

Message 1

baaab aabaa aaaaa aaabb
aaabb aaaaa baabb aaaaa

Message 2

baabb aabaa baaba baabb
abaaa aaabb aabaa aaaaa
baaba

Message 3

ababa aabaa aabaa abbbb
abbab abbba baabb aabaa
baaba

F4. Baconian symbol chart (original synthetic teaching figure)

Answer: _____
_____
_____

31. (200 points) Use Figure F4. Decode Message 2 with the pictured Baconian chart. Cipher groups encode letters. The plaintext starts with TEST; restore word spaces.

Refer to F4. Baconian symbol chart (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (200 points) Use Figure F4. Decode Message 3 with the pictured Baconian chart. Cipher groups encode letters. The plaintext starts with KEEP; restore word spaces.

Refer to F4. Baconian symbol chart (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____